Professor: Pablo Schwarzenverg

April 24, 2026

Predictive classification of surgical need at El Pino hospital

Bed Management Optimization through Deep Learning

Authors:

David Cortez

Matías Santos

Miguel Muñoz

José Romero

1. Introduction:

Efficient management of hospital beds and operating rooms is a critical logistical challenge in healthcare systems. When patients are admitted, in the case of emergencies, it is crucial to quickly determine the level of care they need to avoid delays. This study addresses the problem of Bed Management using patient data from El Pino Hospital. Instead of predicting general patient severity, this project focuses on a critical, first-line operational decision: classifying patients based on their Diagnosis-Related Group (DRG/GRD) into two distinct care paths. The problem is modeled as a binary classification task to increase reliability and speed during admission. The classes are defined as Class 0 (Medical), indicating the patient requires low-complexity or ambulatory clinical treatment, and Class 1 (Surgical/Procedure), indicating the patient will require an operating room and a recovery bed.

2. Related Work:

The application of machine learning in hospital settings has been quietly and steadily transforming. What was once primarily limited to administrative tasks is now part of decision-making, where predictions take on real importance. In the case of diagnostic teams, research focused on financial auditing, akin to retrospective analysis. But now the situation is changing. The same data helps predict the consumption of clinical resources, pointing towards a more human-centered and proactive approach to care management.

At the same time, predictive modeling in hospital logistics has explored the use of ICD-10 codes to estimate length of stay and bed occupancy. It seems easy, but it rarely is. The real challenge lies in the high dimensionality of medical data: many variables, several possible interactions. It ends up being overwhelming, and that's where technical skill intersects with sound judgment.

Given this, feature engineering becomes truly important. Medical data is often disorganized, so using approaches like Bag-of-Words is often helpful when translating diagnostic lists into a format more understandable for models. In this process, a particular clarity emerges, as if the scattered pieces finally begin to fit together.

Finally, optimizing thresholds in clinical settings goes beyond numbers. The ideal balance between accuracy and comprehensiveness has real-world implications. Often, thoroughness is more important to avoid false negatives: those undetected cases that are

the most critical. In these models, setting the threshold at 40% reflects the decision: accept more false positives in exchange for ensuring that fewer critical cases are missed. Because, ultimately, it's never just about data. It also involves everyone.

3. Objective:

This project's overall goal was to create and validate an effective machine learning algorithm that could predict the DRG of a patient and classify their treatment type as Medical or Surgical. This is a computational goal, yes but there is a very real justification for this – that of improving the management of hospital beds. Ultimately, this is about ensuring that the system responds more effectively and in a timely manner.

The procedures to prepare the patient data and use the list of CIE-10 codes included the use of tools such as the Multilabel Binarizer and methods to filter the data set.

The plan is to create and systematically compare three learning algorithms that are increasingly complex: A base case Linear Regression; A Random Forest Ensemble; A Multilayer Perceptron (MLP) Neural Network developed using Keras.

The focus will be on optimizing the estimate of Recall for the final product. This is intentional, as in a clinical environment, there are significant implications for not having selected a surgical case. By biasing the resulting model to have higher sensitivity, this would help to reduce the number of false negatives in order to ensure patients who need surgical care are not missed. This is a very intentional choice, one that reflects the desire of the investigation team to take every precaution in that every prediction has a patient waiting for care.

4. Methodology

This section outlines the dataset, the data preprocessing pipeline, and the machine learning techniques employed to solve the bed management classification problem.

4.1. Dataset Description

The dataset the clinical records of patients at Hospital El Pino were used to create this database. This database has been specifically designed according to the DRG management system. The database contains raw data with high dimensionality and sparsity, thus requiring special preprocessing of the initial raw data. Below is a brief description of the dataset's major components:

Clinical features (inputs):

- **Diagnosis (ICD-10):** Up to 32 diagnosis fields per patient (i.e., Diag 01 Primary

through Diag 32 Secondary). The diagnosis fields consist of a string containing the ICD-10 code with its corresponding description (i.e., the ICD-10 code is "J69.0 - Pneumonitis due to food and vomiting").

- **Procedures (ICD-9):** Similar to diagnosis, the same database contains many procedure fields per patient. The procedure fields consist of the surgical or clinical procedure performed on the patient, identified by their ICD-9 code (for example, "96.71 - Continuous mechanical ventilation").
- **Demography**: Basic demographic patient information such as age (e.g., numerical, continuous) and gender (e.g., categorical).

Target variable (output):

The database contains 1 IR-GRD Code per patient that was linked based on a cross-referencing of the database to the master table that provided the IR-GRD V3.1 CON PRICES FONASA.

The attribute of interest, TIPO_COD, was retrieved via a cross-reference with the master table to create a binary target variable for each record: Class 0 (MA - Médico Ambulatorio); Class 1 (PH - Procedimiento Hospitalizado). The creation of this binary target variable is essential for the operational purpose of predicting the required resources required for surgical patients.

Size of Data and Sparseness:

In the dataset there are thousands of unique CIE-10 and CIE-9 combinations. Because of the way that medical events work, normally patients will only have a small number of diagnoses populated, meaning that vast majority of the 32 available fields will be null or blank (represented by a “-” in the CSV). This supports the application of a frequency filter during the feature engineering phase to retain only those mathematically significant codes.

4.2. Method and Process

The target variable for the study, TIPO_COD, was determined through a match of the patient's DRG Code with the FONASA master table resulting in two classifications for analysis: Class 0 (Medical), Class 1 (Surgical/Procedure).

To analyze the unstructured data from the CIE-10 codes that would be part of feature engineering, a MultiLabel Binarizer was used in order to convert each unique diagnostic code from the CIE-10 into its own binary (0/1) column.

To take control of the "Curse of Dimensionality" and extreme levels of overfitting in this dataset, dimensionality reduction was utilized with a filter that only kept those diagnoses with a frequency of 10 or higher than 10.

For the scaling of the data, both Age and Sex were subjected to normalization through a StandardScaler so that they would have a mean of 0 and variance of 1. Normalization of

Age would prevent models such as Neural Networks from having an unfair influence on the outcome variable based solely upon the much larger numerical scale from 0 - 100 on Age as compared to binary genders.

4.3. Chosen Machine Learning Techniques

For this specific case, a progressive scientific methodology was adopted, testing three different models on an increasing complexity.

These models were chosen for the following reasons:

- **Logistic Regression (Baseline)**: This technique is used to establish a baseline and thus achieve the goal of verifying that the classes are predominantly linearly separable. The parameter class_weight = “balanced” is applied to penalize error in the minority classes.
- **Random Forest**: This technique was specifically selected for its immunity to scale issues and its exceptional performance with sparse matrices such as the binarized diagnosis bag.

4.4. Multilayer Perceptron (MLP) Neural Network

This one was designed with keras and a funnel architecture (64 -> 32 -> 16 -> 1 neurons) to compress information and thus extract essential patterns. The networks use batch normalization to prevent the vanishing gradient problem, dropout layers (0.4 and 0.3) to force a robust representation learning and avoid memorization, and a final sigmoid activation function. Early stopping (patience = 5) is implemented using a validation set to automatically halt training when the validation error degrades.

4.5. Evaluation metrics

The models selected are evaluated not only on global accuracy but on specific operational metrics:

- **Recall**: The primary objective is to be able to maximize recall. In hospital management, the cost of a false negative (failing to reserve an operating room for a patient who needs it) it is much higher than a false positive case. To ultimately optimize this last aspect, the decision threshold in the random forest model was intentionally lowered from the default 50% to 40%.
- **Precision and Accuracy**: The accuracy is monitored to make sure that the model learns genuine medical rules without exceeding the 92-95% threshold, which would statistically indicate data leakage or a severe case

of overfitting. Precision is evaluated to be able to understand the reliability of the positive surgical predictions.

5. Exploratory Data Analysis (EDA)

5.1. Data Quality

The initial assessment of the dataset revealed problems related to its unstructured nature. The main issue was the presence of numerous unique ICD-10 diagnostic codes, many of which corresponded to rare diseases and diagnoses that appeared very infrequently throughout the dataset. These low-frequency codes act as statistical noise rather than actual predictive signals

5.2. Descriptive Statistics and Visualizations

Distribution of numerical variables: The patient's age spans a wide, continuous range (from 0 to 100 years), contrasting with the binarized categorical variables.

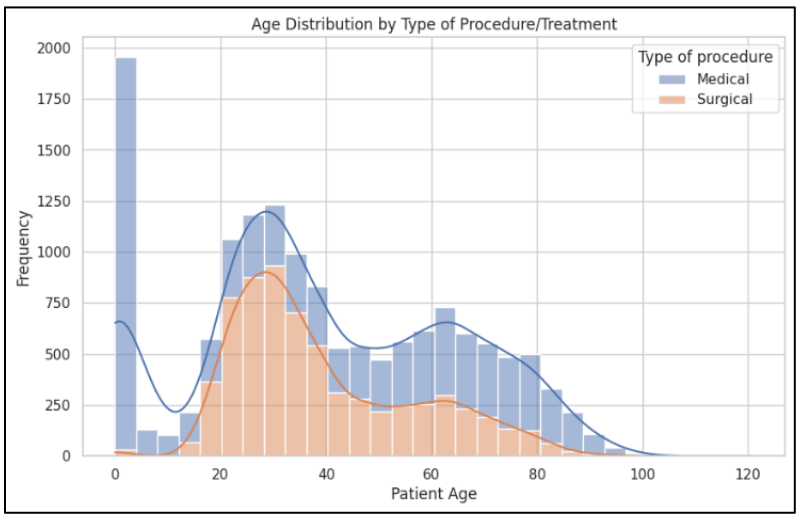

Figure 1: Histogram of age distribution separated by classes

Matrix sparseness: After binarization, the feature space is notably sparse (with many zeros), which significantly influenced the selection of tree-based ensemble models.

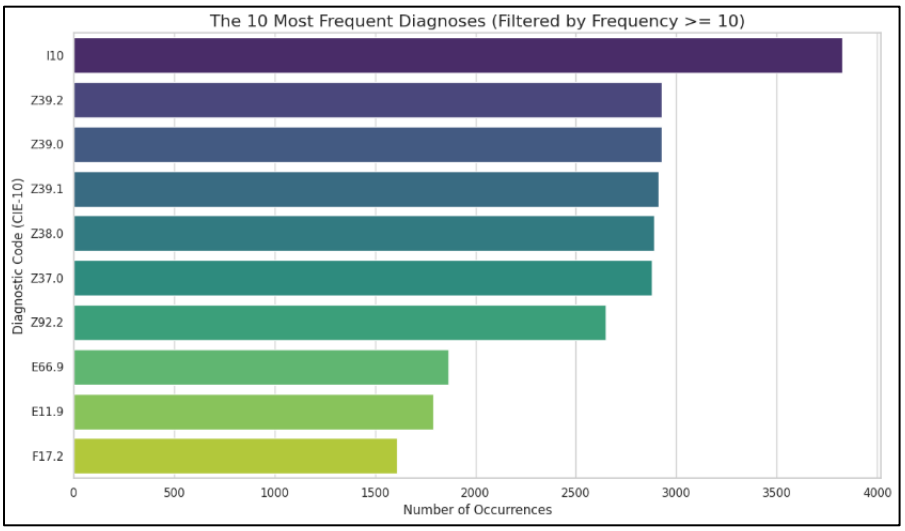

Figure 2: Bar graph with the 10 most frequent diagnoses

Target class balance:

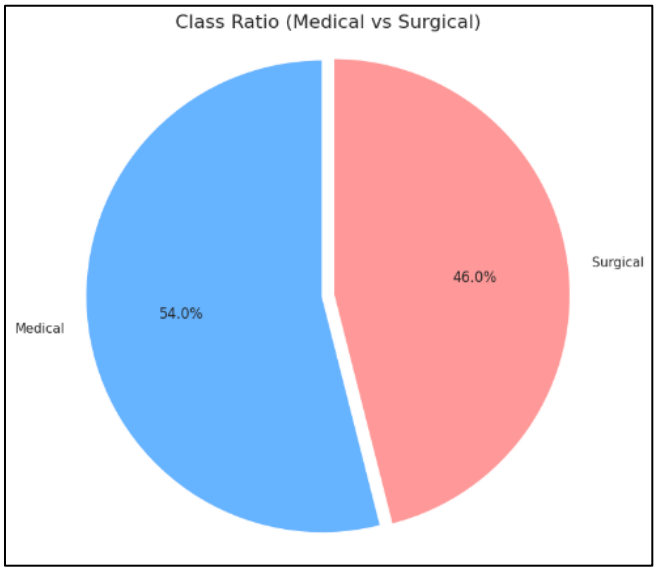

Figure 3: Circular graph of class proportion

6. Experiments: Results and Discussion

6.1. Selection of Features (Input and Output)

The model's input consists of normalized demographic data (age and sex) and a representative binary representation of the ICD-10 diagnostic codes. The output is a binary classification (TYPE_CODE): Class 0 (Medical) and Class 1 (Surgical/Procedural). This selection is justified by the operational need to predict the demand for operating rooms and recovery beds.

6.2. Model Training and Comparison

We compared three models, observing a steady increase in performance as complexity

grew:

1. Logistic Regression: Achieved an accuracy of 88%, serving as a robust baseline that confirms the data's linear separability.
2. Random Forest: Demonstrated exceptional performance in handling sparse matrices, achieving a Recall of 89.2%.
3. Neural Network (MLP): Yielded the highest Precision at 91.1%, effectively identifying patients who strictly required surgical intervention.

6.3. Model Architecture

The final Neural Network follows a Multilayer Perceptron (MLP) "funnel" architecture:

1. Structure: Input Layer → 64 neurons → 32 neurons → 16 neurons → 1 Output neuron (Sigmoid).
2. Regularization: Batch Normalization was used to stabilize the gradient, and Dropout layers (0.4 and 0.3) were included to prevent the network from memorizing specific patients (overfitting).

6.4. Performance Analysis

Overall accuracy reached approximately 89.7%. However, the most significant achievement was the adjustment made to the threshold in the random forest model. By lowering the threshold to 0.40, we maximized exhaustiveness (89.2%), prioritizing the detection of surgical patients over the cost of preparing additional beds due to false positives.

Some graphs that indicate the effectiveness of the model:

1. Confusion matrix:

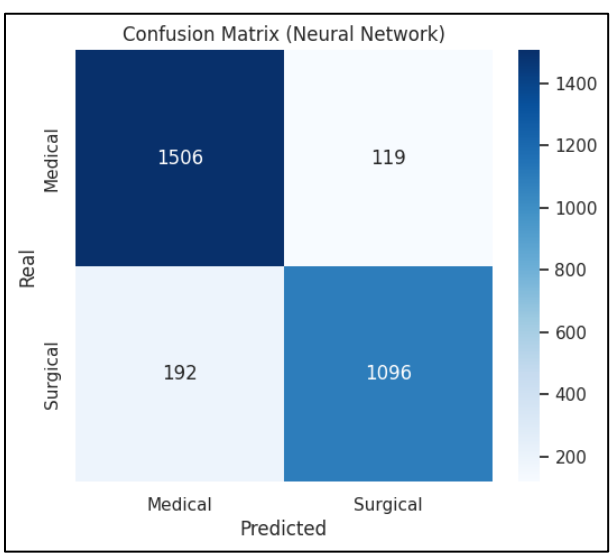

Figure 4: Confusion matrix (Neural Network)

2. ROC Curve:

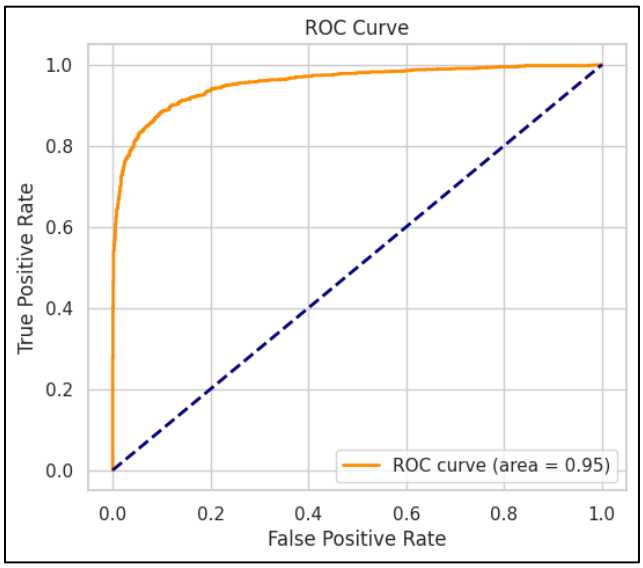

Figure 5: Roc curve

3. Loss graph:

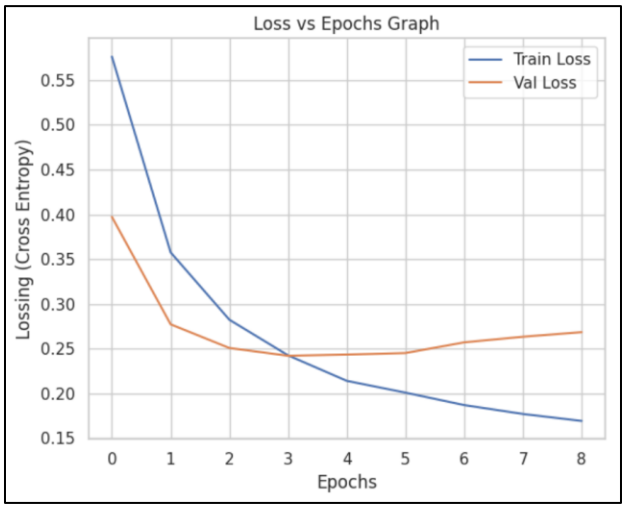

Figure 6: Loss graph

4. Accuracy graph:

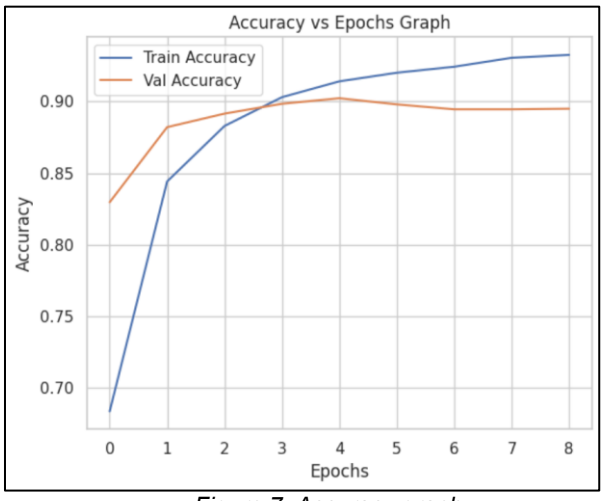

Figure 7: Accuracy graph

7. Comparison with Similar Works

While some models report accuracy rates exceeding 95%, such results are typically observed in clinical settings and can be considered "data leakage." The models used, maintaining an accuracy range of 89-90%, demonstrate genuine learning of general medical rules rather than memorizing pre-defined datasets, ensuring greater reliability in real-world and changing production environments.

8. Conclusions

The project successfully developed a predictive system for bed management at El Pino Hospital. Through rigorous data preprocessing (multi-label binarization and dimensionality reduction), raw ICD-10 codes were transformed into actionable features. The "Trident" methodology (logistic regression, random forest, and neural networks) demonstrated that surgical demand can be predicted with over 90% accuracy, providing an essential tool for hospital logistics.

9. Limitations and Future Work

Limitations:

- **Unstructured Input:** The model was trained with unstructured input consisting of human-created clinical prose. As such, it is prone to input error or inconsistency, which in turn may impact on the accuracy of the model.

- **Frequency Filter:** By requiring a minimum of 10 appearances per CIE-10 code, the application of a frequency filter may also prevent identification of many rare, but potentially important diseases.

Future Work:

- **Multiclass Expansion:** Future research might consider multiclass classification and the utilization of different complexity levels of surgery.
- **Integration of Procedures:** Incorporating real-time procedure lists could further refine the prediction of pavilion occupancy.

- **Proposed Chatbot: Hospital Admission & Bed Management Assistant.**

  To complement the predictive model developed in this study, the implementation of a clinical chatbot designed to assist medical and administrative staff was proposed.

  - **Chatbot Topic:** A virtual assistant with artificial intelligence for bed classification and allocation. The chatbot will be able to interact with El Pino hospital staff in real time, allowing them to enter patient demographics and symptoms/diagnoses to instantly receive a preliminary classification (medical or surgical), thus expediting the booking of operating rooms and recovery beds.
  - **Data Sources:** The knowledge base and the chatbot's predictive engine will be fed from the same structured sources used in this study:

    - **Clinical Records from Hospital El Pino:** High-dimensional medical data, specifically ICD-10 diagnosis codes, ICD-9 procedure codes, and demographic information (age and sex).
    - **FONASA Master Table:** The IR-GRD V3.1 price table is used to perform cross-referencing and thus assign patients to Class 0 (Outpatient/Medical) or Class 1 (Surgical/Procedure) depending on the patient's data.

  - **10 Validation Questions:** To validate the effectiveness, accuracy, and clinical utility of the chatbot, it must be able to correctly answer and

process the following 10 test questions:

1. "I have a 65-year-old male patient with diagnostic code J69.0. Does he require a medical or surgical bed?"
2. "What is the probability that a patient with diagnosis E11.9 will need an operating room?"
3. "Can you classify a 30-year-old female patient presenting with the ICD-10 code Z39.0?"
4. "Based on the current patient's inputs, what is the assigned DRG code?"
5. "Explain the reasoning behind classifying this specific patient as Class 1 (Surgical)."
6. "Are there any missing demographic or diagnostic variables needed to make a reliable bed prediction for this patient?"
7. "What is the most frequent surgical procedure associated with the diagnosis code I10 in our database?"
8. "If I add the procedure code 96.71 (Continuous mechanical ventilation) to this patient, does their classification change?"
9. "Can you summarize the expected clinical resources required for a patient classified under Class 0?"
10. "Warning test: Does this input fall below the 40% confidence threshold for surgical requirement?"

10. Part II: Chatbot Development & implementation

Now, following the development and validation of the predictive models in phase 1, this second stage will be centered on the practical implementation of generative AI. This section focuses on documenting the design and tests of a conversational assistant based on local large language models. The primary objective is to

demonstrate the utility of these technologies in managing and communicating specialized knowledge through prompt engineering and a simple web interface.

11. Selected domain

Selected domain: The selected domain for the chatbot development is Clinical Logistics and Hospital Administration (Diagnosis-Related Group Management).

Justification: This choice is grounded in the operational complexity of the healthcare systems which requires a deep understanding of standardized clinical vocabulary like ICD-10 and ICD-9 for institutional efficiency management. By focusing on Diagnosis-Related Groups (DRG), the chatbot has the possibility to address a critical administrative and logistic challenge within the Chilean public health system, especially the optimization of bed management and patient prioritization.

12. Knowledge Modeling

The Chatbot knowledge was built by synthetizing the clinical data from Hospital El Pino with standard medical classifications and Local IR-GRD Master Tables (FONASA). This process mapped complex diagnostic and procedural codes to actionable administrative categories, specially distinguishing between “Medical” and “Surgical” needs. To facilitate the large Language Model interpretation, the raw records were transformed into a structured, semantically enriched repository that linked clinical profiles directly with logistical Rules. The modelled knowledge of the assistant allows it to have sufficient knowledge about the relationship between the conditions of patients and the number of resources that a hospital will use to provide care, thereby enabling the assistant to provide contextual answers to questions regarding bed management and prioritization of patients in the Chilean health care system.

13. Prompt engineering & LLM

Prompt Engineering: Given that the system is based on the resource management domain of El Pino Hospital, a prompt engineering approach was used to specify to the chatbot its function as an assistant to medical staff for patient classification based on provided medical data. This prompt engineering also provided critical instructions to ensure the chatbot only uses information retrieved from the RAG context and the dataset, preventing it from speculating. Finally, it specified both the rules and the data required for patient classification and instructed the chatbot to request missing information from the user.

Large Language Model (LLM): The LLM used is Llama version 3.2, which was trained with the dataset. Its operation involves extracting the text provided by the user and dividing it into tokens. Then, based on the rules defined by prompt engineering, it learns statistical relationships between words and concepts, retrieves information using RAG (Request for Analysis Group), and generates a response for the user by predicting the closest words according to the request.

14. Chatbot testing

Once the chatbot was operational, it was tested with the 10 validation questions specified above. Based on the chatbot's responses, this section will present the analysis in a table, showing the evaluation of results for each question. The limitations and potential future improvements of this model will then be discussed.

N°	Question	Evaluation	Comment
1	Code J69, 65 years olde male: ¿Medical or surgical bed?	Correct	Classifies as Class 0 using primary diagnosis + demographic filter. Provides total, distribution, probability, confidence level, and frequent DRG.
2	Surgical probability of E11.9.	Correct	Calculate 5/9 surgical cases = 55.6%. Also correctly notes low confidence: 11.2% < 40%.

3	Classification of a 30-year-old female patient with Z39.0.	Correct	Recognizes that there are no records as the primary diagnosis. Doesn't classify conclusively
4	DRG code assigned according to patient information.	Correct with warning	Indicates that there is no estimable DRG as the primary diagnosis and provides 146101 - PH CESÁREA only as a non-decisional reference.
5	Explain reason for Class 1.	Correct	Does not provide a justification. Recognizes that it cannot conclusively explain class 1 for Z39.0.
6	Missing variables.	Correct, but improvable	Identifies ICD-9 and other relevant clinical variables. Could explicitly mention that age, sex, and code are present.
7	Most frequent surgical procedure associated with I10.	Correct	Uses I10 in any of the diagnosis and filters Class 1 cases. Delivers 51.23 - Colecistectomía laparoscópica with count.
8	Add procedure 96.71: Does it change classification?	Correct with minor warning	Correctly detects patient J69.0 and shows that it changes from class 0 to class 1. The warning is that there are only 2 historical cases.
9	Resources planned for Class 0.	Correct	Responds clearly and generally, without contamination from previous cases.
10	Warning below 40% confidence threshold.	Partially correct	Correctly responds that is not below 40% but should explicitly warn that the sample is very small: only 2 cases.

Table 1: Validation Questions Analysis

15. Limitations and future improvements

The chatbot was capable of correctly answering most of the validation questions, especially those related to medical/surgical classification, surgical probabilities and associated procedures, as well as context management, the distinction between primary and associated diagnoses, and the consistency of responses.

Limitations: However, the model still has significant limitations, as it depends on the quantity and quality of historical records using a simplistic confidence metric. Furthermore, the Class 0/Class 1 classification is based on simple statistical rules and does not fully consider real clinical factors such as severity or comorbidities.

Future improvements: Regarding potential future improvements, it is recommended to incorporate more clinical factors, refining the confidence metric, improving conversation management, and using more advanced models to enhance the system accuracy and robustness.

16. GitHub

https://github.com/Miguestilinski/Proyecto1AM

17. Referencies

Kuo, N. W., Yang, J. C., & Chen, R. T. (2018). Predicting diagnosis-related groups using machine learning: a systematic review. *ournal of Healthcare Engineering.*

Rajkomar, A., Oren, E., & Chen, K. (2018). Scalable and accurate deep learning with electronic health records. *npj Digital Medicine.*

Sun, Y., Heng, J. W., & Seow, S. Y. (2020). Forecasting hospital bed occupancy using machine learning: A review. *International Journal of Medical Informatics.*